

Make-Or-Break First Scenario

Make-or-Break is a building simulation game based on physics, enables users to design structures as well as experiment with them against various types of natural disasters. This first scenario will give the fundamental building and physics features that create the bases of our game experience. Players will have access to basic blocks, and basic physic forces, in order to see how their structures react to these conditions. You can run the game by either double clicking on the game executable file or starting the game through the Unity Editor's project window

Scenario "Build and Test Structure"

Upon launching the game, you will see the main menu screen displaying these options:

1. Play mode
2. Creative mode
3. Settings
4. Quit

The logo for Make-Or-Break should be present on the main menu screen as well and the overall appearance of the game should be based around a simplistic style to represent construction.

To enter the game, the user must click on "Play mode" (for ease of development, "Creative mode" will be focused on first in order to get the physics and building correctly). When they do, the system will then switch to the BuildMode scene. This scene contains an empty 3D space with a flat surface upon which the player can build your structures. The camera view is at its default position which allows you to see the entire building area.

To move around the environment, the user can use their keyboard (WASD) for camera movement and mouse movement for rotating the camera to see the various angles of the construction site. The camera motion should be designed to be smooth and responsive so that the user may move freely throughout the environment without any delays or interruptions to their movement.

On the left side of the screen is an easy-to-read, simple building materials panel that lists out the available building materials (WOODEN BEAM, CONCRETE BLOCK, CONCRETE BLOCK (REBAR), WINDOW FRAME, PLASTER WALL, TREES, etc.). Each of these building materials is represented by a button that can be clicked on with an icon or text label next to it.

In order to select a building material, the user clicks on "CONCRETE BLOCK" from the building materials panel. When a building material is selected, the building material is highlighted, and the cursor changes to indicate that the user is in the placing mode.

After selecting a building material, the user clicks the grid in the construction area to place the building material. The system will then place the concrete block in the nearest snapping grid location on the grid. The user continues to click to place as many blocks as required for a simple foundation of four blocks by four blocks using the construction materials. The minimum foundation required is a 4x4 block of concrete.

After the player has selected that they want to use "WOODEN BEAM", they will then place multiple beams down on top of the concrete foundation in order to begin creating walls for a basic structure. These beams will align with the concrete block below them so that it is simple to build vertically.

Once the player has created a simple structure (which will consist of approximately 8-10 different building components), the player will click on the "START TEST" option that can be found in the upper right corner of the screen.

At this point, there will be a transition from the building mode to the testing mode. The UI will change to show the user that they have now entered TEST MODE and give them one option for the natural disaster, something like: EARTHQUAKE (MAGNITUDE 5.0). The player must click the "APPLY NATURAL DISASTER" button.

Once the player has clicked on apply natural disaster, Unity's physics engine will turn on and each block that the player has created will be given a Rigidbody component if it does not already have one. The simulated ground forces will apply periodical ground shaking to each of the structures by applying periodical forces to the ground plane and then to each of the structures that are in contact with it. Each structure will react to the applied forces according to Unity physics calculations.

The player will then see their structure sway and possibly collapse depending upon the strength of their design. The blocks will fall, beams will break, and the structure will react to gravity and collision forces realistically. All of the items will respond to appropriate collision detection. Blocks will not pass through each other and will stack naturally and fall naturally.

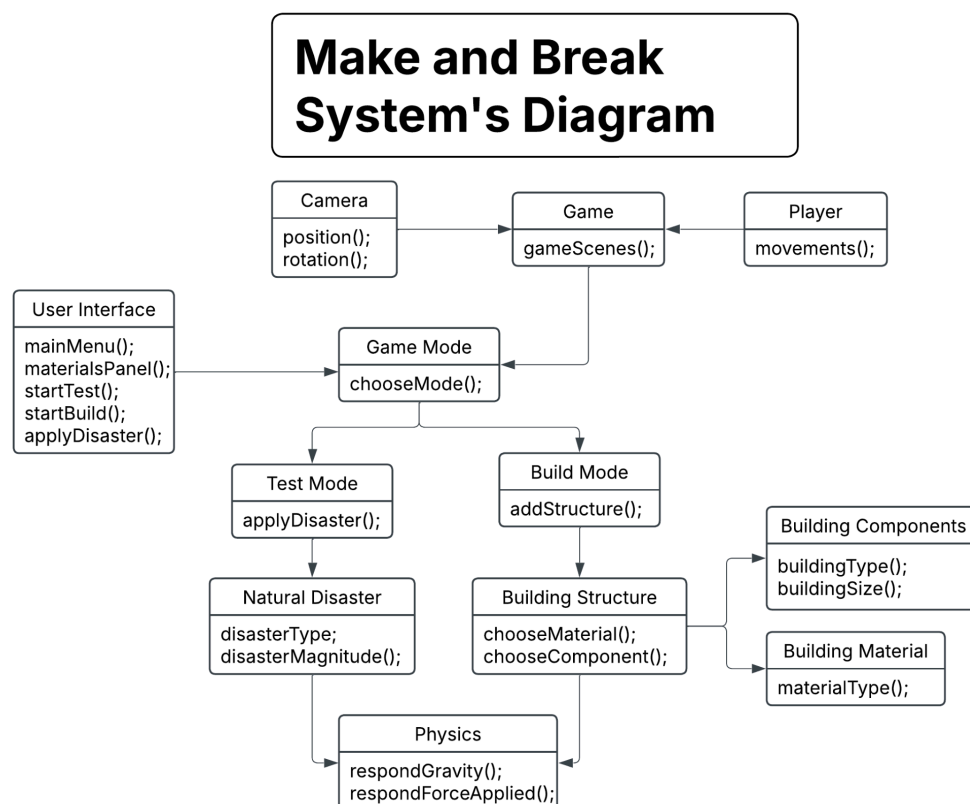


Figure 1. System Diagram showcasing Unity project structure